

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of B. Sc. (Biological Sciences) Backlog Examination, September 2019

Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 18/09/2019 Day: Wednesday Time: 10:00 A.M. To 10:30 A.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	The following method is used for detection of specific DNA fragments on a nitrocellulose membrane by hybridization to a labelled DNA probe.		01
	(i) Northern blotting	(ii) Southern blotting	
	(iii) Eastern blotting	(iv) Western blotting	
2.	The following electromagnetic radiations have wavelengths longer than visible light.		01
	(i) X-rays	(ii) UV-rays	
	(iii) Infrared radiation	(iv) All of them	
3.	In gel-filtration chromatography, the _____ will be excluded from the pores of the gel beads and will therefore migrate at a faster rate in the chromatography column.		01
	(i) Intermediate size analytes	(ii) Small analytes	
	(iii) Large analytes	(iv) All of them	
4.	In _____ elution process a single solvent is used as the mobile phase for chromatographic separation of analytes.		01
	(i) Gradient	(ii) Isocratic	
	(iii) Isopycnic	(iv) All of them	

5.	The purity ratio of $A_{260}/A_{280} = 2.6$ was obtained for a genomic DNA sample isolated from bacteria, this absorbance ratio indicates contamination of _____ in the purified DNA.		01
	(i) Chloroform	(ii) RNA	
	(iii) Proteins	(iv) All of them	
6.	The following reagent is used to inhibit the activity of nucleases during isolation of bacterial genomic DNA.		01
	(i) Tris-Cl	(ii) Ethidium bromide	
	(iii) EDTA	(iv) Isopropanol	
7.	Which of the following is used as a tracking dye to observe the migration of DNA samples in agarose gel electrophoresis.		01
	(i) Ethidium bromide	(ii) Bromophenol blue	
	(iii) Diethylpyrocarbonate	(iv) Blue dextran	
8.	The _____ chromatography separates biomolecules based on highly specific biological interactions (such as antigen-antibody or enzyme-substrate interactions).		01
	(i) Ion-exchange	(ii) Size-exclusion	
	(iii) Affinity	(iv) All of them	
9.	In centrifugation, the term RPM is used to describe _____.		01
	(i) Revolutions per milliseconds	(ii) Relative centrifugal field	
	(iii) Revolutions per minute	(iv) All of them	
10.	Which of the following thermostable enzyme is used for the polymerase chain reaction (PCR).		01
	(i) Lysozyme	(ii) Proteinase K	
	(iii) Taq DNA polymerase	(iv) RNase	
11.	The _____ chromatography is used to separate biomolecules based on their net-surface charge.		01
	(i) Gel-filtration	(ii) Affinity	
	(iii) Ion-exchange	(iv) All of them	
12.	In SDS-PAGE, the following reagent is used to denature and confer a net negative charge on proteins.		01
	(i) Sodium dodecyl sulfate	(ii) Ammonium per sulfate	
	(iii) β -mercaptoethanol	(iv) Phenol-chloroform	
13.	According to Beer-Lambert law the absorbance of a solution is directly proportional to its _____.		01
	(i) Concentration	(ii) Reduction	
	(iii) Denaturation	(iv) Precipitation	
14.	The following fluorescent protein was isolated from <i>Aequorea victoria</i> (Jelly fish).		01
	(i) Insulin	(ii) DsRed	

	(iii) GFP	(iv) Antibody	
15.	Of the following, which microscope has best resolution and magnification?		01
	(i) Phase contrast microscope	(ii) Fluorescence microscope	
	(iii) Light microscope	(iv) Electron microscope	
16.	The following spectroscopy is also known as vibrational spectroscopy?		01
	(i) Atomic emission spectroscopy	(ii) Infrared spectroscopy	
	(iii) Atomic absorption spectroscopy	(iv) UV-Visible spectroscopy	
17.	Which of the following microscope is most suitable to visualize live cells without performing any staining or sample preparation?		01
	(i) Scanning electron microscope	(ii) Transmission electron microscope	
	(iii) Phase contrast microscope	(iv) Fluorescence microscope	
18.	In protein purification, when the ammonium sulfate concentration is increased the solubility of proteins decreases promoting their precipitation, this phenomenon is known as _____.		01
	(i) Desalting	(ii) Dialysis	
	(iii) Salting in	(iv) Salting out	
19.	The _____ microscope utilizes electromagnetic lenses as objectives.		01
	(i) Fluorescence	(ii) Dark-field	
	(iii) Electron	(iv) All of them	
20.	The atomic absorption spectroscopy is based on the principle of _____.		01
	(i) Absorption of radiation by non-excited atoms	(ii) Molecular vibration	
	(iii) Fluorescence	(iv) All of them	

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Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 18/09/2019 Day: Wednesday Time: 10:30 A.M. To 1:00 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any four).		20
1.	Describe the various components of a gas chromatography system.	05
2.	Describe different components and stages of a typical polymerase chain reaction (PCR).	05
3.	Explain different types of paper chromatography methods.	05
4.	Explain the principle of the following: (a) Size-exclusion chromatography, (b) Affinity chromatography	05
5.	Explain the procedure used for the isolation of genomic DNA from bacteria.	05
6.	Explain in detail the southern blotting method.	05
SECTION - II		
Q-III Answer the following questions as directed (answer any six).		30
1.	Explain the SDS-PAGE method for separation of proteins.	05
2.	Explain the principle and instrumentation of IR spectroscopy.	05
3.	Explain the procedure of plasmid DNA isolation by alkaline-denaturation method.	05
4.	Give an overview of various types of centrifuges used in biological centrifugation.	05
5.	Explain the instrumentation of a UV-visible spectrophotometer.	05
6.	Give an overview of the different steps involved in purification of proteins.	05
7.	Explain the major components of an electron microscope.	05
8.	Explain the principle and instrumentation of a fluorescence microscope.	05